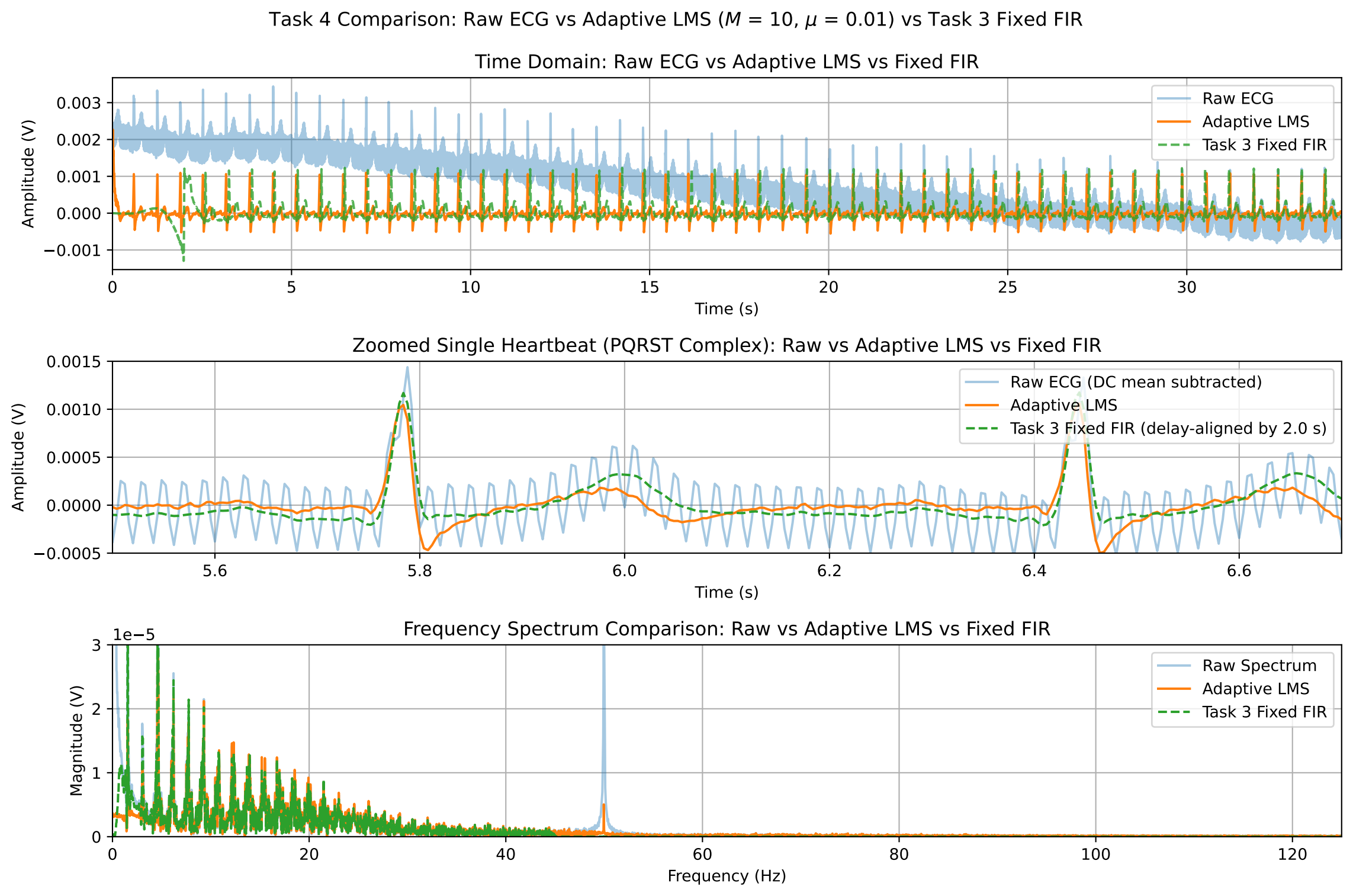

Task 4 Comparison: Raw ECG vs Adaptive LMS (M = 10, μ = 0.01) vs Task 3 Fixed FIR
Time Domain: Raw ECG vs Adaptive LMS vs Fixed FIR
Raw ECG
Adaptive LMS
Task 3 Fixed FIR
Amplitude (V)
Time (s)
Zoomed Single Heartbeat (PQRST Complex): Raw vs Adaptive LMS vs Fixed FIR
Raw ECG (DC mean subtracted)
Adaptive LMS
Task 3 Fixed FIR (delay-aligned by 2.0 s)
Amplitude (V)
Time (s)
Frequency Spectrum Comparison: Raw vs Adaptive LMS vs Fixed FIR
1e−5
Raw Spectrum
Adaptive LMS
Task 3 Fixed FIR
Magnitude (V)
Frequency (Hz)